Cwikel's Plus-Minus Isometry Conjecture for Banach Lattices

Run `fa_banach_001`, agent `agent_lane_04`

2026-06-23

Status

Candidate full result, likely valid. This packet proves the isometry conjectured in Remark 4.1 of Cwikel's paper [3]: for a couple of complexified Banach lattices of measurable functions on the same measure space, Cwikel's continuous plus-minus norm agrees with Calderon's complex interpolation norm.

The earlier order-continuous endpoint result has been moved into the packet's `history/` folder.

Source Problem

The source paper is Michael Cwikel, "Some alternative definitions for the 'plus-minus' interpolation spaces $\langle A_0, A_1\rangle_\theta$ of Jaak Peetre," arXiv:1502.00986v2 [3]. The relevant statement is Remark 4.1 on page 13.

> "periodic" norms on $[A_0, A_1]_\theta$.
>
> *Remark* 4.1. The inclusion $\langle A_0, A_1\rangle_\theta \subset [A_0, A_1]_\theta$ is the easy part of the proof of another interesting property of the space $\langle A_0, A_1\rangle_\theta$, namely that it *coincides* with $[A_0, A_1]_\theta$, to within equivalence of norms, whenever (A_0, A_1) is a couple of complexified Banach lattices of measurable functions on the same underlying measure space. This property has sometimes proved useful, for example in [6], and I hope to use it further in the very near future, in a forthcoming paper (which in fact has been the main motivation for me to write this note). I also conjecture that this coincidence to within equivalence of norms of $\langle A_0, A_1\rangle_\theta$ and $[A_0, A_1]_\theta$ for couples of lattices is even an *isometry* when $\langle A_0, A_1\rangle_\theta$ is equipped with the norm of $\langle A_0, A_1\rangle_{\theta,(0)}$, i.e., the norm associated with the apparently new "continuous" method for its construction presented in Section 3. Hence my interest in explicitly noting that the embedding $\langle A_0, A_1\rangle_{\theta,(r)} \subset [A_0, A_1]_\theta$ has norm

The remark recalls that, for couples of complexified Banach lattices of measurable functions on the same measure space, the plus-minus space $\langle A_0, A_1\rangle_\theta$ is known to coincide with Calderon's complex interpolation space $[A_0, A_1]_\theta$ up to equivalent norms. It then conjectures that the coincidence is actually isometric when the plus-minus side is equipped with the continuous norm $\langle A_0, A_1\rangle_{\theta,(0)}$.

Idea of the Proof

Cwikel already proves the norm-one inclusion

$$\|x\|_{[X_0,X_1]_\theta} \le \|x\|_{\langle X_0,X_1\rangle_{\theta,(0)}} .$$

Thus only the reverse inequality is needed.

The direct Calderon-product proof works immediately for elements whose pointwise Calderon factorization has bounded logarithmic ratio: put a small scalar kernel at that logarithmic ratio and the endpoint plus-minus estimates are dominated by the two Calderon factors. The old order-continuous packet removed unbounded logarithmic-ratio tails by order continuity.

The full proof uses a sharper observation. Calderon's complex space for Banach lattices is not the whole upper Calderon product in general; by Shestakov's theorem it is the closure of $X_0 \cap X_1$ in the Calderon product norm [4, 5]. For elements already in $X_0 \cap X_1$, the unbounded ratio tails can be killed without order continuity: on the positive tail, the X_1 norm decays exponentially while the X_0 norm is bounded; placing that tail at the midpoint $t = K/2$ makes both weighted endpoint norms small. The negative tail is symmetric. Therefore the norm-one estimate first holds on $X_0 \cap X_1$, and then passes to its Calderon-product closure, which is the complex interpolation space.

Notation and Background

Let X_0 and X_1 be complex Banach lattices of measurable functions on a common measure space, and let $0 < \theta < 1$. We write $\langle X_0, X_1\rangle_{\theta,(0)}$ for Cwikel's continuous plus-minus space and norm from [3, Section 3]. Thus $x \in \langle X_0, X_1\rangle_{\theta,(0)}$ if $x = \int_{-\infty}^{\infty} u(t)\, dt$ for an admissible locally piecewise continuous $X_0 \cap X_1$-valued function u, and

$$\|u\|_J = \max_{j=0,1} \sup_{\phi} \left\| \int_{-\infty}^{\infty} e^{(j-\theta)t} \phi(t) u(t)\, dt \right\|_{X_j},$$

where the supremum is over locally piecewise continuous scalar ϕ with $\|\phi\|_\infty \le 1$. The plus-minus norm is the infimum of $\|u\|_J$ over all such representations of x.

We use the Calderon product gauge

$$\|f\|_{X_0^{1-\theta} X_1^{\theta}} = \inf M,$$

where the infimum is over all measurable $b \ge |f|$ and real measurable s such that

$$be^{-\theta s} \in X_0, \quad be^{(1-\theta)s} \in X_1, \quad \left\|be^{-\theta s}\right\|_{X_0} \le M, \quad \left\|be^{(1-\theta)s}\right\|_{X_1} \le M.$$

This is the usual Calderon product norm, after balancing the two endpoint factors.

We shall use two standard facts. First, Calderon's embedding is contractive:

$$\|x\|_{X_0^{1-\theta} X_1^{\theta}} \le \|x\|_{[X_0,X_1]_\theta}.$$

Second, Shestakov's Banach-lattice theorem says that Calderon's complex interpolation space is the closure of $X_0 \cap X_1$ in the Calderon product norm:

$$[X_0, X_1]_\theta = \overline{X_0 \cap X_1}^{\,\|\cdot\|_{X_0^{1-\theta} X_1^{\theta}}}.$$

Yuan's note [5, p. 2] records this classical theorem and its relation to the inner complex method.

Two Elementary Plus-Minus Estimates

Lemma 1 (One-point representation)**.** *For every $z \in X_0 \cap X_1$ and every $a \in \mathbb{R}$,*

$$\|z\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \max\{e^{-\theta a} \|z\|_{X_0}, e^{(1-\theta)a} \|z\|_{X_1}\}.$$

Proof. Let k_δ be the normalized indicator of $[-\delta/2, \delta/2]$ and put $u_\delta(t) = z\, k_\delta(t-a)$. Then $\int u_\delta(t)\,dt = z$. For every test function ϕ with $|\phi| \le 1$,

$$\left\| \int e^{-\theta t}\phi(t)u_\delta(t)\,dt \right\|_{X_0} \le C_{0,\delta} e^{-\theta a} \|z\|_{X_0},$$

and

$$\left\| \int e^{(1-\theta)t}\phi(t)u_\delta(t)\,dt \right\|_{X_1} \le C_{1,\delta} e^{(1-\theta)a} \|z\|_{X_1},$$

where $C_{0,\delta} = \int e^{-\theta r}k_\delta(r)\,dr$ and $C_{1,\delta} = \int e^{(1-\theta)r}k_\delta(r)\,dr$. Both constants tend to 1 as $\delta \downarrow 0$. □

Lemma 2 (Bounded logarithmic-ratio bands)**.** *Let g be a measurable function. Suppose that for some measurable $b \ge |g|$ and some real measurable s taking values in a bounded interval,*

$$y_0 := be^{-\theta s} \in X_0, \qquad y_1 := be^{(1-\theta)s} \in X_1.$$

Then

$$\|g\|_{\langle X_0, X_1\rangle_{\theta,(0)}} \le \max\{\|y_0\|_{X_0}, \|y_1\|_{X_1}\}.$$

Proof. First suppose that s takes only finitely many values. Put $\alpha = g/b$ on $\{b>0\}$ and $\alpha = 0$ on $\{b=0\}$, so $|\alpha| \le 1$ and $\alpha b = g$. Let k_δ be as in the previous lemma and set

$$u_\delta(t,\omega) = \alpha(\omega)b(\omega)k_\delta(t - s(\omega)).$$

Because s has finite range, u_δ is locally piecewise constant as an $X_0 \cap X_1$-valued function and has compact t-support. Also $\int u_\delta = g$. For every admissible ϕ,

$$\left| \int e^{-\theta t}\phi(t)u_\delta(t)\,dt \right| \le C_{0,\delta} y_0, \qquad \left| \int e^{(1-\theta)t}\phi(t)u_\delta(t)\,dt \right| \le C_{1,\delta} y_1.$$

The lattice property gives the corresponding endpoint norm estimates, and letting $\delta \downarrow 0$ proves the finite-valued case.

If s is merely bounded, choose finite-valued measurable q with $|q - s| \le \eta$. Applying the finite-valued case with q in place of s yields

$$\|g\|_{\langle X_0, X_1\rangle_{\theta,(0)}} \le e^{\eta} \max\{\|y_0\|_{X_0}, \|y_1\|_{X_1}\}.$$

Let $\eta \downarrow 0$. □

Norm-One Estimate on the Intersection

Lemma 3. *For every $h \in X_0 \cap X_1$,*

$$\|h\|_{\langle X_0, X_1\rangle_{\theta,(0)}} \le \|h\|_{X_0^{1-\theta}X_1^{\theta}}.$$

Proof. Fix $M > \|h\|_{X_0^{1-\theta}X_1^{\theta}}$. Choose $b \ge |h|$ and measurable real s such that

$$y_0 = be^{-\theta s} \in X_0, \qquad y_1 = be^{(1-\theta)s} \in X_1,$$

and $\|y_0\|_{X_0}, \|y_1\|_{X_1} \le M$. For $K > 0$ put $h_K = h\mathbf{1}_{\{|s| \le K\}}$. Lemma 2 gives

$$\|h_K\|_{\langle X_0, X_1\rangle_{\theta,(0)}} \le M.$$

We claim that $h_K \to h$ in the plus-minus norm. On the positive tail $p_K = h\mathbf{1}_{\{s>K\}}$, we have

$$\|p_K\|_{X_0} \le \|h\|_{X_0}, \qquad \|p_K\|_{X_1} \le e^{-(1-\theta)K} \|y_1\|_{X_1} \le e^{-(1-\theta)K} M.$$

Applying Lemma 1 with $a = K/2$ gives

$$\|p_K\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \max\{e^{-\theta K/2} \|h\|_{X_0}, e^{-(1-\theta)K/2} M\} \to 0.$$

On the negative tail $n_K = h\mathbf{1}_{\{s<-K\}}$, we similarly have

$$\|n_K\|_{X_0} \le e^{-\theta K} M, \qquad \|n_K\|_{X_1} \le \|h\|_{X_1}.$$

Lemma 1 with $a = -K/2$ gives

$$\|n_K\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \max\{e^{-\theta K/2} M, e^{-(1-\theta)K/2} \|h\|_{X_1}\} \to 0.$$

Thus $h_K \to h$ in $\langle X_0, X_1\rangle_{\theta,(0)}$. Since all h_K have plus-minus norm at most M, completeness of Cwikel's continuous plus-minus space [3, Section 5] gives $\|h\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le M$. Letting $M \downarrow \|h\|_{X_0^{1-\theta}X_1^{\theta}}$ proves the lemma. $\square$

The Full Isometry

Theorem 1. *Let X_0 and X_1 be complex Banach lattices of measurable functions on the same measure space. Then, for every $0 < \theta < 1$,*

$$\langle X_0, X_1\rangle_{\theta,(0)} = [X_0, X_1]_\theta$$

isometrically.

Proof. Cwikel's Theorem 4.3 gives the contractive embedding

$$\|x\|_{[X_0,X_1]_\theta} \le \|x\|_{\langle X_0,X_1\rangle_{\theta,(0)}}.$$

It remains to prove $\|x\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \|x\|_{[X_0,X_1]_\theta}$.

Let $x \in [X_0, X_1]_\theta$ and fix $M > \|x\|_{[X_0,X_1]_\theta}$. By Calderon's contractive embedding, $\|x\|_{X_0^{1-\theta}X_1^{\theta}} \le \|x\|_{[X_0,X_1]_\theta}$. By Shestakov's theorem, choose $h_m \in X_0 \cap X_1$ such that $\|h_m - x\|_{X_0^{1-\theta}X_1^{\theta}} \to 0$ and, after discarding finitely many terms, $\|h_m\|_{X_0^{1-\theta}X_1^{\theta}} \le M$ for all m.

Lemma 3 gives

$$\|h_m - h_n\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \|h_m - h_n\|_{X_0^{1-\theta}X_1^{\theta}} \to 0,$$

so (h_m) is Cauchy in $\langle X_0, X_1\rangle_{\theta,(0)}$. Let y be its plus-minus limit. The contractive embeddings

$$\langle X_0, X_1\rangle_{\theta,(0)} \hookrightarrow [X_0, X_1]_\theta \hookrightarrow X_0^{1-\theta}X_1^{\theta}$$

imply that $h_m \to y$ in the Calderon product norm. Since also $h_m \to x$ in that norm, $y = x$ as a measurable function. Therefore $x \in \langle X_0, X_1\rangle_{\theta,(0)}$, and

$$\|x\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \limsup_m \|h_m\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \limsup_m \|h_m\|_{X_0^{1-\theta}X_1^{\theta}} \le M.$$

Letting $M \downarrow \|x\|_{[X_0,X_1]_\theta}$ gives the reverse contractive embedding. Together with Cwikel's inequality, this proves equality of the two norms. $\square$

Verification Notes

The proof separates the only delicate non-order-continuous issue from the lattice-complex interpolation theorem. The weighted ℓ_∞ model shows why the whole upper Calderon product cannot be used directly: unbounded logarithmic-ratio tails need not vanish in sup norm. Shestakov's theorem supplies exactly the missing regularity: Calderon's complex space is the closure of $X_0 \cap X_1$ inside the Calderon product. On $X_0 \cap X_1$ the tails are controlled by placing them at midpoint parameters and using their actual endpoint membership.

Bounded novelty/literature check performed on 2026-06-23: the run indexes were searched for `1502.00986`, `plus-minus`, `Peetre`, `Calderon`, `lattice`, and isometry keywords. Web searches for exact phrases from Remark 4.1 and for "plus-minus interpolation Banach lattices complex interpolation isometry" did not reveal a later exact resolution. The proof uses Shestakov's known identification of complex interpolation with the regular Calderon-product closure, as summarized in Yuan [5], but the norm-one plus-minus estimate supplied here appears not to be stated in the located sources.

Recommended verifier focus: check Lemma 3, especially the positive/negative tail estimates in the plus-minus norm, and confirm that the cited Shestakov closure theorem applies to the same class of complexified Banach lattices intended in Cwikel's Remark 4.1.

References

[1] A. P. Calderon, Intermediate spaces and interpolation, the complex method, *Studia Math.* 24 (1964), 113–190.

[2] M. Cwikel and N. J. Kalton, Interpolation of compact operators by the methods of Calderon and Gustavsson-Peetre, arXiv:math/9210206, 1992.

[3] M. Cwikel, Some alternative definitions for the "plus-minus" interpolation spaces $\langle A_0, A_1 \rangle_\theta$ of Jaak Peetre, arXiv:1502.00986v2, 2015.

[4] V. A. Shestakov, On complex interpolation of Banach spaces of measurable functions, *Sibirsk. Mat. Zh.* 15 (1974), 923–933.

[5] W. Yuan, A note on complex interpolation and Calderon product of quasi-Banach spaces, arXiv:1405.5735, 2014.